Position-based prediction with coupled deformable contact

Numerical qualification, parameter fitting and reserved cross-interaction evaluation

We fit stiffness and friction from surface positions during a squeeze, shear and release, then predict new interactions using nonlinear deformation, rigid-body motion and explicit contact. Contact forces are computed from the scene; observed force labels are not inputs.

Fixed primary comparison

The primary group contains 24 reserved forecasts across eight objects: slide, curved, moving_sphere. Membership was fixed before fresh outcomes. All declared cases remain, including numerical-reference failures and failed predictions.

Primary method	Complete	World RMSE (mm)	Shape RMSE (mm)
Inferred nonlinear FEM	24/24	5.155	0.973
Rigid + shared contact	24/24	6.823	0.975
Adapted PhysTwin spring core	16/24	8.704	0.970
Position ridge	24/24	45.466	4.414
True-parameter coarse FEM	24/24	3.338	0.837

World error measures positions in the shared scene. Shape error removes a best-fitting proper rigid transformation per frame. Means for incomplete methods are conditional on completion; the coverage column is part of the result. Repeated interactions share eight objects. Across all 24 primary cases, world error is 5.155023 mm for FEM versus 6.823473 mm for rigid contact; aligned shape error is 0.973114 versus 0.975289 mm, a difference of 0.002174 mm. World-motion gains must not be interpreted as evidence of a substantial deformation-prediction gain.

Joint calibration/forecast numerical checks pass for 22/24 primary cases. This coverage does not change the primary denominator.

Against rigid contact, inferred FEM wins 20, loses 4 and ties 0 across 24 completed pairs; mean paired error reduction is 1.668 mm. Descriptive 95% object-cluster bootstrap interval for the mean paired reduction: [0.217, 2.697] mm. The 10,000 resamples draw eight objects with replacement, keeping repeated interactions together. 8 objects contribute completed pairs; 10000 resamples contain paired scores. The interval is conditional on paired completion and is not a population guarantee.

Matched primary control	Complete	Paired	World RMSE (mm)	FEM gain (mm)
Fixed nominal FEM	24/24	24	5.073	-0.082
Vertex-contact formulation	24/24	24	6.118	0.963

Vertex contact is refit on the same observations and matches scene, mass and mesh. Its damping mass differs from the surface formulation, so this does not isolate quadrature alone.

Method and information available

1. Observe and fit

The context contains 18 surface nodes with 0.1 mm Gaussian position noise and known tool poses. A fixed grid of 7 stiffness values and 5 friction values is scored by calibration position error. Selection never uses future target positions. Initial geometry, density, gravity, Poisson ratio, viscosity and collider geometry are known.

2. Predict the coupled scene

The deformable object uses compressible Neo-Hookean tetrahedral FEM, objective viscous damping and consistent P1 mass. Rigid bodies have separate position and momentum states and exact uniform-volume inertia. Contact applies equal and opposite reactions. Future commanded tool positions drive the rollout; the future position of a free rigid body is predicted.

Exact piecewise-linear normal-force integration over clipped rest-surface triangles for planes and boxes, refined friction quadrature, and conservative sparse material-surface quadrature for spheres; shared configuration between reference and all physical predictors. Exact normal integration does not imply exact friction, continuous collision detection or converged trajectories.

3. Compare matched explanations

Rigid contact fits friction while suppressing deformation. The oracle diagnostic receives true stiffness and friction but retains the coarse predictor. Fixed nominal FEM uses 6000 Pa and friction 0.35. Position regressions are weaker controls without scene collision geometry. The native baseline uses the authors' unchanged PhysTwin spring-force core within this study's graph, masses, contact, integration and global parameter search. It is not the full authors' method. Actual calibration-library coverage is nonlinear FEM: 35/35 valid, 0 failed; adapted PhysTwin spring core: 2/35 valid, 33 failed; rigid contact: 5/5 valid, 0 failed. Failed candidates remain in the accounting. These counts describe the matched adaptation used here. Native primary forecasts complete for 16/24; its mean is conditional on those cases and must not be directly contrasted with a different FEM cohort. On the 16 completed matched pairs, FEM wins 13, loses 3 and ties 0; the mean paired world-error reduction is 3.022124 mm. Native stress forecasts complete for 0/8.

Parameter recovery is a separate outcome

FEM parameter fits complete for 8/8 reserved objects. Mean absolute relative stiffness error: 33.571%. Mean absolute friction error: 0.112. These are finite-grid point estimates, not a posterior or calibrated uncertainty. Fine/coarse model discrepancy can bias effective parameter estimates even when motion improves.

Transfer scope

The reserved materials are new combinations inside the declared parameter range. Forecast interactions differ from the calibration interaction, but their families were used during development. This is not unseen-geometry, unseen-material-range or real-world generalization.

Numerical evidence and retained stress cases

The common predictor has 4 cells per axis and a 12.5 microsecond step. Reference choices were frozen from development evidence.

Scenario	Reference cells / axis	Companion cells / axis	Base step (us)
squeeze	8	10	25.0
slide	8	10	25.0
curved	8	10	25.0
moving_sphere	12	10	25.0
edge_drop (stress)	8	12	12.5

All 150 declared reference jobs are accounted for: a primary, a half-step run and a distinct-mesh companion for every object and scenario. Spatial comparisons use common 27-point and 2678-surface-point probes. Fixed gates are 1 mm world-coordinate RMS, 0.5 mm aligned shape RMS, 1 mm body-center RMS, 0.25 mm temporal RMS, at most 1 mm penetration and positive element volume with $J > 0.02$. Primary qualification requires time-step checks at both development resolutions.

These are two-resolution consistency checks, not a proof of continuum convergence or measured physical truth. Per-case flags are retained, and missing or failed comparisons are unqualified.

Drop diagnosis

The drop family remains development-unqualified and is excluded from the fixed primary scope by a prospective decision. All 8 reserved drop forecasts remain in the full 32-case secondary accounting. An individual fresh numerical pass does not promote a stress case into the primary group. No reliable-drop claim is made. Fresh joint calibration/forecast numerical checks pass for 0/8 stress cases, compared with 22/24 primary cases. These flags do not alter either fixed denominator.

The original sharp support showed substantial mesh sensitivity. A rounded capsule rail was tested as a separately declared engineering redesign with the same width and top height. Its underside differs from the box; it is not merely isolated corner smoothing. Contact integration and time resolution were checked separately from spatial resolution. All failed trials are retained.

Software and preservation

Independent verification reconstructs observations, selection losses, metrics, fixed cohort membership and reference quality. Frozen sources, protocol, compiler artifacts and evidence digests make the selected experiment reviewable. The 405 recorded prior artifacts remain unchanged.

What this establishes and what remains

The concrete contribution of this campaign is a more carefully qualified coupled simulation comparison: material-surface contact, consistent inertia, position-only fitting, explicit free-body response and honest accounting of failed references. Prediction gains are measured against matched controls on the fixed primary group.

Public benchmark transfer

This controlled synthetic extension does not establish improvement on both archived public benchmarking datasets. The separate development-only rigid-pose transfer pilot is not promoted: SoftVT mean gain 0.0214mm with10 wins/10 losses; PhysTwin is unchanged.

The 21-case development pilot changed only volume-based rigid pose/inertia in an existing release predictor. It did not transfer the full coupled deformation model. Exact old replay and unchanged observed prefixes were independently checked. A broader public-dataset adapter still needs compatible geometry, units, interaction inputs and its own numerical validation.

Remaining scientific limits

Known scene geometry and gravity are substantial inputs. Unknown support geometry, arbitrary boundary fields, multiple deformable bodies, image reconstruction, calibrated uncertainty and uncertainty-driven probing are not established by this experiment. The adapted native spring comparison cannot support an official PhysTwin or conference benchmark ranking.

Videos and reproducibility

Three video cases were fixed before outcome inspection. They share world coordinates and camera motion, display method failures and numerical-reference status, and use no deformation exaggeration. Playback is four times slower than physical time. Stress clips are visibly labeled.

Code: scripts/coupled_contact_v13. Frozen protocol, cohort tables, numerical checks and video provenance: reports/coupled-surface-v13. Historical study: reports/coupled-contact-v13. Raw trajectories, datasets and checkpoints remain local.

Read the fixed primary table together with completion counts, matched paired comparisons and numerical qualification coverage. The full and stress tables remain available; no failed case was silently removed to improve the headline result.